

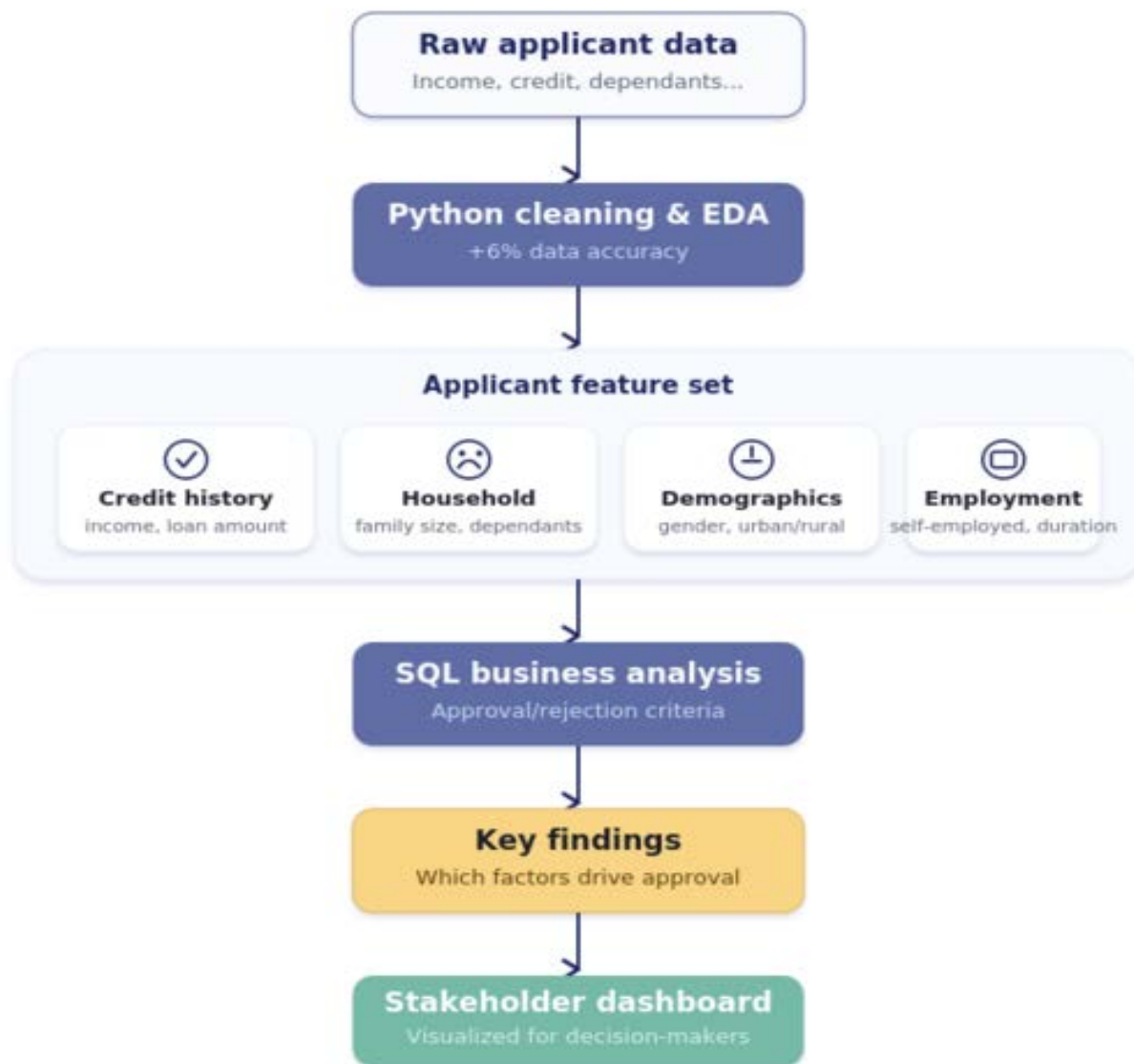
Loan Approval Prediction Analysis

Data Analysis Project — Python, SQL & Stakeholder Dashboarding

Chirag Seervi
Data Analyst / AI QA

Project Pipeline

How the project moved from raw applicant records to a stakeholder-ready view of loan approval drivers.



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+6% Data accuracy gain from cleaning & augmentation	10 Applicant features analyzed across the dataset	3 Deliverables — cleaned dataset, SQL insights, dashboard
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Overview

Analyzed historical loan-company data to understand how real-world applicant circumstances shape loan approval and rejection outcomes. The project moved from raw, messy records to a stakeholder-ready view of what actually drives a lending decision, combining Python for data preparation, SQL for business-question analysis, and a dashboard for non-technical stakeholders.

Methodology

- **Python (cleaning & EDA):** handled missing values, corrected inconsistent entries, and augmented the dataset — improving overall data accuracy by 6%.
- **SQL (business analysis):** queried the cleaned data to answer real-world scenario questions and surface which criteria most influence approval or rejection.
- **Dashboard:** visualized the findings so stakeholders could review approval patterns without touching raw data.

Applicant Features Analyzed

Credit history	Income
Loan amount	Loan duration
Family size	Dependants
Graduate status	Self-employed status
Gender	Urban / rural location

Impact

The cleaned, feature-rich dataset and the resulting SQL-driven analysis gave stakeholders a clear, evidence-based view of which applicant characteristics correlate most strongly with loan approval — turning a raw applicant record set into a decision-support tool.

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Key Findings

- 1. 79% vs 18%** — Applicants with a credit history are approved ~79% of the time, versus only ~18% for those without one — the single strongest predictor of approval.
- 2. +16%** — Applicants with high income have a ~16% higher chance of approval than lower-income applicants.
- 3. +62%** — Low-risk applicants (good income paired with a low loan amount) have ~62% more approval chance than high-risk applicants.
- 4. Top rejection cause** — Lack of credit history is the leading cause of rejection, followed by low income and a high requested loan amount.
- 5. +10%** — Graduates have a ~10% higher approval chance than non-graduates.
- 6. +35%** — Small loans are approved ~35% more often than high-amount loans.

Business Recommendations

- **Prioritize applicants with an established credit history** — it is the strongest single signal of repayment likelihood (79% vs 18% approval).
- **Weight income favorably in scoring** — higher-income applicants warrant a moderate approval boost (~16%).
- **Score applicants on a combined risk profile** (income + loan amount) rather than either factor alone — low-risk profiles merit streamlined approval given their 62% edge.
- **Use the top three rejection drivers as triage criteria:** missing credit history, low income, and high loan amount. Flag applicants with two or more of these for manual review rather than automatic rejection.
- **Apply a small positive weighting for graduate status** (~10%) as a secondary, not primary, criterion.
- **Tier the approval path by loan size** — route smaller loan requests through a lighter-touch approval process, and apply added scrutiny or collateral/guarantor requirements to larger loan amounts.

Bottom line: prioritize applicants who have a credit history, stable or higher income, and a smaller requested loan amount relative to that income. Applicants missing credit history, with low income, or requesting large loan amounts should move to enhanced review rather than an automatic decision.